

Reproduction Report: Exact-Geometry Cut-Cell Nitsche Infrastructure, Verification, and a Preregistered Protocol for Bounded Stabilization under Sliver Cuts

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August 25, 2026

Abstract

This document is a *reproduction and verification report*, not a novel-research paper: it records the construction, verification, and first measured exercise of a computational instrument, together with the frozen protocol under which its main experiments will be judged. The subject is symmetric Nitsche methods on unfitted meshes, whose penalty λ_T under the classical volume-scaled choice, $\lambda_T = C_k \rho(T)/h_T$ with $\rho(T) = |T \cap \partial\Omega|/|T \cap \Omega|$, diverges when the physical domain occupies a vanishing fraction of a background element (a *sliver cut*), inflating stiffness-matrix condition numbers and destroying accuracy through round-off. The report contributes three things. First, an exact-geometry computational framework in which circle cuts are clipped in closed form via Green’s theorem — carrying no tessellation error at any cut fraction — ellipses reduce affinely to the same exact path, and only genuinely non-quadric level sets fall back to adaptive subdivision with per-cell *measured* quadrature error. Second, a preregistered protocol (formulas

1 Introduction

Let $\Omega \subset \mathbb{R}^d$ be embedded in a background domain D with a mesh $\mathcal{T}_h$ independent of $\partial\Omega$. For the Poisson problem $-\Delta u = f$ in Ω , $u = g$ on $\Gamma = \partial\Omega$, the symmetric Nitsche form seeks $u_h \in V_h$ with

$$a_h(u_h, v_h) = \sum_T \int_{T \cap \Omega} \nabla u_h \nabla v_h - \sum_T \int_{\Gamma_T} [(\partial_n u_h) v_h + u_h (\partial_n v_h)] + \sum_T \lambda_T \int_{\Gamma_T} u_h v_h = \sum_T \int_{T \cap \Omega} f v_h + \int_{\Gamma_T} g (\lambda_T v_h - \partial_n v_h) \quad (1)$$

where $\Gamma_T = T \cap \Gamma$. Discrete coercivity requires, through local trace-inverse estimates, a penalty scaling like $\lambda_T \geq C \rho(T) h_T^{-1}$ with $\rho(T) = |\Gamma_T|/|T \cap \Omega|$ (Cui and Kanschat, 2025). Under a sliver cut, $|T \cap \Omega| \rightarrow 0$ while $|\Gamma_T|$ stays bounded below, so $\rho(T)$ and

Green’s theorem: each directed line piece contributes $\frac{1}{2}(a_x b_y - b_x a_y)$, and an arc swept from angle a_0 to a_1 about centre c contributes

$$\frac{R}{2} \left[c_x (\sin a_1 - \sin a_0) - c_y (\cos a_1 - \cos a_0) + R(a_1 - a_0) \right], \quad (2)$$

with signed sweep taken in the traversal direction. Boundary measures $|\Gamma_T|$ are arc lengths, likewise exact. Interior integrals use a Gauss rule on the polygonal fan plus tensor Gauss strips over each crescent in a (θ, ρ) frame bounded below by the chord; angular panels exist purely for quadrature order, so combined mass reproduces the closed-form area to relative 10^{-13} while every basis-product integral converges exponentially. Because a circle may cross a convex element up to six times, the loop construction pairs every exit crossing with the next entering crossing counter-clockwise and asserts all boundary arcs subtend less than π .

Ellipse cuts ride the same exact path: the affine map $y = D^{-1}(x - c)$ sends the ellipse to the unit disk and the element to another triangle; weights pick up $|\det D|$, normals pull back through D^{-T} , and the physical boundary measure integrates directly against

originating methodology document are logged inside the preregistration itself (PETSc unavailability, supplementary mass-normalized eigenvalue logging, deferred dual-backend cross-check, and the subdivision-quadrature policy). Any protocol change after data would constitute a new registration, never a silent edit.

4 Experiments

4.1 Verification suite

Thirty-three automated checks pass. The sliver family is validated against closed-form circular segments: placing the chord at depth d makes $|T \cap \Omega| = \frac{1}{2}(\theta - \sin \theta)$ and $|\Gamma_T| = \theta$, $\theta = 2 \arccos d$, and the computed quantities match to 10^{-12} relative down to $d = 0.999999$ (cut fraction $\sim 5 \times 10^{-7}$). Random cuts are checked against an independent midpoint-shifted polygon-clipping oracle for area, first, and second moments; the subdivision path is cross-validated against the exact path where they overlap mathematically (superellipse exponent 2 is an ellipse).

This verification campaign earned its cost by catching defects that mass-consistency checks alone do not reveal: quadrature

at every mesh (shift-invert Lanczos returned positive smallest eigenvalues with explicit convergence flags). Two honest observations. First, even these “well-cut” configurations contain moderately small cuts — the circle grazes some cells — which is precisely why raw $\kappa(A_h)$ for $k = 2$ is large and non-monotone between $n = 32$ and $n = 64$: raw smallest eigenvalues are basis-dependent, which is why the preregistration logs mass-normalized companions alongside the gate quantity. Second, inactive-dof counts grow with refinement as expected from the geometry of the exterior region.

4.3 What is not yet measured

No controlled-sliver, random-ensemble, curvature, sign-changing, or three-dimensional results exist at the time of writing; tiers T1–T5 and the falsification gates H1–H3 therefore appear in this manuscript only as registered protocol. We decline to extrapolate how any bounded formula will behave at $\varepsilon \leq 10^{-3}$: the point of the frozen gates is that the data will speak first.

5 Related work</

github.io/nitsche-cutcell-methodology/; reproducibility repository (source, tests, run logs) <https://github.com/mohamedhossammohamed/nitsche-cutcell-methodology>. Preregistration SHA-256 b5f108ebad0465e7599a39726b301210946bc6198fc7054dfba615652108721f (2026-08-25T01:54:08Z).

7 Conclusion

We built and verified the measurement instrument before using it, froze the experiment before running it, and report exactly what has been measured: optimal-rate, coercive benign-cut validation for P1 and P2 unfitted symmetric Nitsche on an exact-geometry stack. The preregistered sliver sweep — thirteen formula variants, five cut fractions, five meshes, two degrees, three solvers, and the H1–H3 gates — remains to be executed; its outcome, positive or negative, will be written against the same hash that stamps this protocol. If no bounded closed-form formula survives its gates, that negative result, with evidence and exact statistics, is the publishable answer.

References